

Introduction to Modeling and Simulation

History of Modeling and Simulation (Dubois, 2018)

- Modeling and simulation originated from the concept of numerical simulation in 1940, which was also marked as the beginning year of computer science.
- World War II (WWII) stimulated researches in numerical simulations.
- The **Electronic Numerical Integrator and Computer (ENIAC)** was the first electronic programmable computer that could solve a large class of numerical problems.
- The first numerical simulation is often associated with the *Manhattan Project* of the United States in partnership with the United Kingdom and Canada, which led to creating of the first atomic bomb during WWII.
- The first models that were simulated by *John Von Neumann* and his colleagues represented the nuclear detonation process.
- In the mid-1950s, *Keith Douglas Tocher* established the **three-phase method**, widely used for discrete simulation. Tocher used this method to develop a model to simulate the operations of a production factory, integrating the flow of raw materials and different stocks.
- In the late 1950s, **Computer-Aided Design (CAD)** was developed. This allowed engineers to convert drawn designs into numerical designs and visualize in 2D and 3D. Then, several companies in the transportation industry started to adopt CAD.
- In 1967, the first *Winter Simulation Conference* was held. This world conference aimed to present new insights to and from different industries, solely focusing on simulation.
- From 1980 to 1995, computers become more accessible and simulation was greatly developed. Modeling and simulation were then utilized by companies in diverse fields to address various issues such as software consolidation and visual animation.
- From 1996 to the present, numerous innovations, such as the Internet and virtual reality, were developed to support the growth and expansion of modeling and simulation, making it more profitable for any industry.

Advantages and Disadvantages of Modeling and Simulation (BBC, 2021)

Advantages:

- Modeling and simulation through computing devices are safer, more efficient, and effective than an actual or real-life simulation.
- It can be used to identify unexpected problems.

- Modeling and simulation allow stakeholders to test a product or a system before building it.
- It can be used to explore and evaluate possibilities related to the product or system.
- Results are generally accurate.

Disadvantages:

- It is possible to commit errors in setting-up the model or programming the simulation that would greatly affect the results.
- A higher initial investment cost is required to build a simulation model and run multiple simulations.
- It requires some time to articulate results.

General Concepts in Modeling and Simulation

- A **system** is an organized collection of entities linked together according to a plan to accomplish a specific goal (Tutorialspoint, n.d.). Note that entities of a system for one study can only be a subset of another larger system. The state of a system can be defined as an assemblage of variables necessary to describe a system at a particular instant of time with respect to the study's goals.
- **System limitations** must always be clearly identified. These basically defines the composition of a system, inside and out. These are observer-dependent, time-dependent, and system-dependent. Thus, *demarcation* is required to differentiate entities in an environment (Chaturvedi, 2010).
- **System components** are the fundamental parts of a system having an organized relationship with each other or the environment. It is possible to present system components in the form of differential equations (Chaturvedi, 2010).

Basic System Components:

- **Entity** – It refers to an object of interest in a system.
- **Attribute** – It refers to the property of an entity.
- **Event** – It is an instantaneous occurrence that may change the state of the system.
- **System State** – It is a collection of variables necessary to describe a system at a given point in time, relative to the study's objectives.
- **System Progress** – The gradual development of a system can be analyzed by monitoring the changes in a system's state.

- The **system environment** is the "supersystem" within which a smaller system or organization operates. It also determines how a system must function.
- **System Analysis** is the process of collecting and interpreting facts and identifying problems and possible solutions, possibly through system decomposition. It is conducted to study systems or parts of the system in order to rationalize a set of objectives. It is a problem-solving technique that improves a system and ensures the efficiency and effectiveness of all the components in achieving their purpose (Tutorialspoint, n.d.).

Classifications of Systems (Chaturvedi, 2010)

According to Time Frame:

- Discrete system** – The state variables change instantaneously at separated points in time. Examples are queuing systems such as bank systems, telephone networks, and traffic lights.
- Continuous system** – The state variables change continuously with respect to time. Examples are the solar system, the movement of an airplane, and charging a battery.
- Hybrid system** – It is a combination of a continuous and discrete system that has the benefit of encompassing a larger class of systems within its structure, allowing more flexibility in terms of modeling. An example is continuous traffic along a road with multiple traffic lights.

According to Complexity:

- Physical system** – The components or variables of this system are quantitative and measured through physical devices. Examples are the electrical system, mechanical system, and computer system.
- Conceptual system** – The components or variables of this system are theoretical and in qualitative form. Thus, the quantity of interest cannot be measured. Examples are psychological systems and economic systems.
- Esoteric system** – According to Merriam-Webster dictionary, *esoteric* means requiring or exhibiting knowledge restricted to a small group. This system is intended to be understood by a small number of people with specialized knowledge or interest. Thus, the complexity of this type of system is of the highest order.

According to Degree of Interactions:

- Independent systems** – The events within the system do not affect each other (i.e., event A does not affect event B and vice versa).

- Cascaded systems** – The effects of the events are unilateral (i.e., event A affects event B, B affects C, C affects D, and not in reverse).
- Coupled systems** – The events mutually affect each other (i.e., event A affects event B and vice versa).

According to Uncertainties Involved:

- Deterministic system** – There are no uncertainties in any variable involved in the system.
- Stochastic system** – Some of the variables in the system are arbitrary or inconsistent.
- Fuzzy system** – The variables are difficult to understand and explain clearly.

- A **model** is an examinable representation of a system (Dubois, 2018). It may represent an event and/or an entity, which can be real or contrived. It can depict a system at some point of abstraction or multiple levels of abstraction to represent the system in a mathematically reliable method. The following should always be considered in choosing a simulation model:

1. The function of the system;
2. The characteristics of the system; and
3. The objectives of the study.

- A **simulation** is an applied methodology that can describe the behavior of a system using either a mathematical model or a symbolic model. Simulation is the imitation of an operation of a real-world process, specifically a system, over time (Sokolowski & Bank, 2009). Simulation can also be considered as a predictive tool, which is a profitable investment.

Basic Applications of Simulation

- Gathering insights
- Validation of experiments
- Simulation-based training
- Supports different statistical analysis
- Computer animation
- Controlling real-time processes
- Predicting outcomes
- Evaluating new systems
- Enhanced education

Best Practices in Modeling and Simulation (Dubois, 2018)

- Always define a set of objectives.
 - Include sufficient and necessary physical phenomena in formulating the actual problem.
 - Focus on converting the actual problem into an equation.
 - Configure the model carefully based on the formulated equation.
 - Select the most appropriate software for the implementation of the model.
 - Be mindful of the possible errors during the simulation.
 - Closely monitor the validity level of the simulation results.
 - Identify a proper maintenance frequency for the model.
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- **Aleatory Variable** – These are specific phenomena and events in any environment which are considered random, simply because there is no better way of characterizing them (Sokolowski & Bank, 2009). It practically pertains to the natural randomness in a simulation process. Examples: In flipping a coin, the air current, the irregularities in the landing surface, and the force exerted in flipping the coin are considered aleatory uncertainties.
 - **Epistemic Uncertainty** – These are the scientific uncertainties in the simulation model, commonly caused by limited data and knowledge. It can be characterized by alternative models. The presence of any significant degree of epistemic uncertainty almost always suggests that the simulation will be used for gaining insight (Sokolowski & Bank, 2009). Example: In flipping a coin, the question "Is the coin fair?" can be considered an epistemic uncertainty.

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